Junyong Bae, Ph.D.

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EDUCATION

Ph.D. in Nuclear Engineering (Feb. 2025)

- Ulsan National Institute of Science and Technology (UNIST), Ulsan, Korea
- Dissertation: Deep Learning-based Optimized Simulation and Consequence Estimation with Neural Network Uncertainty for Dynamic Risk Assessment of Nuclear Power Plants
- Advisor: Prof. Seung Jun Lee

B.S. in Nuclear Science and Engineering (Feb. 2019)

- UNIST, Ulsan, Korea
- Minor in Electrical Engineering
- Graduated with Cum Laude

PROFESSIONAL EXPERIENCE

Postdoctoral Researcher (Mar. 2026 – Present)

- Risk Assessment Research Division, Korea Atomic Energy Research Institute (KAERI)
- Advisor: Hyeonmin Kim, Ph.D.
- 지능형 사고 시나리오 예측 및 대응 기술 개발 (가상원자로 개발 사업)

Postdoctoral Researcher (Mar. 2025 – Mar. 2026)

- Nuclear Safety Assessment and Plant HMI Evolution Lab, UNIST
- Advisor: Prof. Seung Jun Lee
- Cyber-attack detection system with deep-learning models (가동원전 안전성 향상 사업)
- AI-supported dynamic PSA project (AI+S&T 사업)

RESEARCH TOPICS

AI Applications for NPP Operations

- Data-driven prediction of plant status and operator support under abnormal, emergency, and severe accident conditions
- High-level autonomous operation via reinforcement learning
- AI-assisted accident diagnosis
- Deep learning-based false data injection detection
- Explainable AI for human–AI interaction

AI-Assisted Dynamic PRA

- Simulation optimization algorithms for massive accident scenarios
- Machine learning-based scenario interpretation and event sequence generation
- Data-driven surrogate models for accident analysis codes
- Risk assessment of dynamic events (e.g., LOOP, FLEX strategies, passive systems)

AI Transformation (AX)

- LLM applications for technical documentation
- Big data analytics for on-site operations

HONORS & AWARDS

- Best Paper Award, *Nuclear Engineering and Technology (NET)*, Korean Nuclear Society, (2026)
- Grand Prize, Best Oral Presentation Awards, ASRAM 2025 (2025)
- Outstanding Doctoral Dissertation Award (2025)

- Best Paper Award, 2024 KNS Fall Conference (2025)
- Best Student Paper Award, ISOFIC 2024 Conference (2024)
- Best Paper Award, 2023 KNS Fall Conference (2024)
- KNS Doosan Enerbility Scholarship (2023)
- KNS Park Chang-kyu PSA Student Paper Award (2022)
- Excellence Award, Nuclear-AI Convergence Case Study Presentation (2021)
- Best Paper Award, 2018 KNS Student Conference (2018)
- National Scholarship for Science and Engineering (2017–2018)

PUBLICATIONS

1. **Junyong Bae**, Seung Jun Lee, "Explainable AI for NPP Parameter Trend Prediction based on Example-based Explanation." ***In preparation.***
2. **Junyong Bae**, Yunyeong Heo, Seung Jun Lee, "Dynamic Risk Quantification of Station Blackout Using Limit Surface Identification and Over-Time Reliability Data." ***In preparation.***
3. **Junyong Bae**, Ji Hyeon Shin, Seung Jun Lee, " False Data Injection Detection in Nuclear Power Plants using Autoencoder with Latent-Conditioned Mahalanobis Scoring." *Nuclear Engineering and Technology,* **Under review**.
4. Donghee Jung, Junyong Bae, Hyun Gook Kang, Seung Jun Lee, "Latent Trajectory Forecasting for Long-Horizon Multivariate Nuclear Power Plant Parameter Prediction." *Energies* 19(18) (2026). https://doi.org/10.3390/en19133048
5. Kyeong Seo Kim, **Junyong Bae**, Wooseok Jo, Dong-San Kim, Seung Jun Lee, "Efficient Simulation of Massive Scenario Sets via Redundant Timeline Identification and Reuse." *Progress in Nuclear Energy* 198, (2026). https://doi.org/10.1016/j.pnucene.2026.106451
6. **Junyong Bae**, Kyeong Seo Kim, Seung Jun Lee, "Two-Stage Deep Learning Model for Nuclear Power Plant Parameter Trend Prediction with Epistemic Uncertainty Quantification." *Applied Soft Computing* 186, 114233 (2026). https://doi.org/10.1016/j.asoc.2025.114233
7. Ji Hyeon Shin, **Junyong Bae**, Seung Jun Lee, "An Approach to Validating Abnormal Event Diagnoses with Explainable AI." *Progress in Nuclear Energy* 189, 105939 (2025). https://doi.org/10.1016/j.pnucene.2025.105939
8. Wooseok Jo, **Junyong Bae**, Dong-San Kim, Seung Jun Lee, "Automatic Accident Sequence Generation using Optimized Simulations for Dynamic Probabilistic Safety Assessment." *Reliability Engineering & System Safety* 266 (2025). https://doi.org/10.1016/j.ress.2025.111643
9. **Junyong Bae**, Seung Jun Lee, "Current Progress in the Application of Artificial Intelligence for Nuclear Power Plant Operation." *Korean Journal of Chemical Engineering* 41 (2024). https://doi.org/10.1007/s11814-024-00246-7
10. **Junyong Bae**, Jae Min Kim, Seung Jun Lee, "Deep Reinforcement Learning for a Multi-Objective Operation in a Nuclear Power Plant." *Nuclear Engineering and Technology* 55 (2023). https://doi.org/10.1016/j.net.2023.06.009 **[Best Paper Award]**
11. Jae Min Kim, **Junyong Bae**, Seung Jun Lee, "Strategy to Coordinate Actions through a Plant Parameter Prediction Model during Startup Operation of a Nuclear Power Plant." *Nuclear Engineering and Technology* 55 (2023). https://doi.org/10.1016/j.net.2022.11.012
12. Ji Hyeon Shin, **Junyong Bae**, Jae Min Kim, Seung Jun Lee, "An Interpretable Convolutional Neural Network for Nuclear Power Plant Abnormal Events." *Applied Soft Computing* 132, 109792 (2023). https://doi.org/10.1016/j.asoc.2022.109792
13. Jeeyea Ahn, **Junyong Bae**, Byung Joo Min, Seung Jun Lee, "Operation Validation

System to Prevent Human Errors in Nuclear Power Plants." *Nuclear Engineering and Design* 397, 111949 (2022). https://doi.org/10.1016/j.nucengdes.2022.111949

14. **Junyong Bae**, Jong Woo Park, Seung Jun Lee, "Limit Surface/States Searching Algorithm with a Deep Neural Network and Monte Carlo Dropout for Nuclear Power Plant Safety Assessment." *Applied Soft Computing* 124, 109007 (2022). https://doi.org/10.1016/j.asoc.2022.109007
15. **Junyong Bae**, Geunhee Kim, Seung Jun Lee, "Real-Time Prediction of Nuclear Power Plant Parameter Trends following Operator Actions." *Expert Systems with Applications* 186, 115848 (2021). https://doi.org/10.1016/j.eswa.2021.115848
16. **Junyong Bae**, Jeeyea Ahn, Seung Jun Lee, "Comparison of Multi-layer Perceptron and Long Short-Term Memory for Plant Parameter Trend Prediction." *Nuclear Technology* 206 (2020). https://doi.org/10.1080/00295450.2019.1693215

SELECTED CONFERENCE PRESENTATIONS

1. **Junyong Bae**, Seung Jun Lee, ”Feature Space Memory Banks for Example-Based Model Explanation in Safety-Critical Applications,” ESREL 2026, Braga, Portugal (Jun. 2026)
2. **Junyong Bae**, Seung Jun Lee, "Example-based Explanation Framework for NPP Parameter Trend Prediction using Latent Space Similarity," KNS 2025 Fall, Changwon, Korea (Oct. 2025).
3. **Junyong Bae**, Wooseok Jo, Kyeong Seo Kim, Seung Jun Lee, "Dynamic Risk Assessment Framework with Limit Surface Search and Success and Failure Domains Analysis," ASRAM 2025, Pattaya, Thailand (Aug. 2025). **[Grand Prize, Best Oral Presentation Award]**
4. **Junyong Bae**, Seung Jun Lee, "Epistemic Uncertainty in Deep Learning Models and Its Application for Model Performance Improvement," ANS 2025 Annual Conference, Chicago, U.S. (Jun. 2025).
5. **Junyong Bae**, Wooseok Jo, Kyeong Seo Kim, Seung Jun Lee, "Parameters Trend Prediction using Two-Stage Deep Learning Models to Capture their Predictive Uncertainty under Optimized Simulations," ISOFIC 2024, Jeju, Korea (Dec. 2024). **[Best Student Paper Award]**
6. **Junyong Bae**, Seung Jun Lee, "Epistemic Uncertainty in Deep Learning Models and Its Application to Simulation Optimization and Consequence Parameter Trend Predictions for Dynamic Risk Assessment," KNS 2024 Fall, Changwon, Korea (Oct. 2024). **[Best Paper Award]**
7. **Junyong Bae**, Ji Hyeon Shin, Seung Jun Lee, "Deep Autoencoding Gaussian Mixture Model for Detection of Manipulation Attacks on NPP Process Data," PSAM 17 & ASRAM 2024, Sendai, Japan (Oct. 2024).
8. **Junyong Bae**, Yunyeong Heo, Wooseok Jo, Seung Jun Lee, "Postprocessing of Multi-dimensional Limit Surfaces Searched by a Deep Learning-based Algorithm," KNS 2023 Fall, Gyeongju, Korea (Oct. 2023). **[Best Paper Award]**
9. **Junyong Bae**, Donghee Jung, Seung Jun Lee, "NPP Emergency Response Planning utilizing Tree Search Algorithm and Deep Learning Models," ESREL 2023, Southampton, United Kingdom (Sep. 2023).
10. **Junyong Bae**, Seung Jun Lee, "Sensitivity Study about Hyperparameters of a Guided Simulation Algorithm with Deep Learning," ASRAM 2022, Daejeon, Korea (Nov. 2022).
11. **Junyong Bae**, Jong Woo Park, Seung Jun Lee, "Deep Learning for Guided Simulation of Scenarios for Dynamic Probabilistic Risk Assessment," PSAM 16, Honolulu, U.S. (Jun. 2022).
12. **Junyong Bae**, Jong Woo Park, Seung Jun Lee, "Uncertainty-aware Limit Surface

Search Algorithm using Deep Neural Network," 2021 ANS Winter Meeting, Washington D.C., U.S. (Dec. 2021).

13. **Junyong Bae**, Jong Woo Park, Seung Jun Lee, "Limit Surface Search Algorithm with Artificial Neural Network and Monte Carlo Dropout Uncertainty Quantification," PSA 2021, Online (Nov. 2021).
14. **Junyong Bae**, Jong Woo Park, Seung Jun Lee, "Adaptive Sampling for Limit Surface Search using Deep Neural Network and Monte Carlo Dropout," KNS 2021 Fall, Changwon, Korea (Oct. 2021). **[KNS Park Chang-kyu PSA Student Paper Award]**
15. **Junyong Bae**, Seung Jun Lee, "Nuclear Power Plant Parameter Prediction Strategy for Human Error Detection," 2019 ANS Winter Meeting, Washington D.C., U.S. (Nov. 2019).
16. **Junyong Bae**, Seung Jun Lee, "Operation Validation System through Time Series Forecasting by Artificial Neural Network," 2018 KNS Student Conference, Yeosu, Korea (Oct. 2018). **[Best Paper Award]**

TEACHING EXPERIENCE

- Teaching Assistant, *Nuclear Safety and Safety Culture*, UNIST (2022 – 2024)
- Teaching Assistant, *NPP Accident Diagnosis using AI Techniques*, UNIST (2022)
- Teaching Assistant, *Introduction to Nuclear Reactor Operation*, UNIST (2021)
- Teaching Assistant, *NPP I&C Systems*, UNIST (2019 – 2020)
- Undergraduate Teaching Assistant, *General Physics II*, UNIST (2017, 2018)
- Project Staff**,** 원자력안전 사전실습 교육강화 사업 (Korea Foundation of Nuclear Safety) (2022-2026)

PROFESSIONAL SERVICE

Journal Reviewer

- Engineering Applications of Artificial Intelligence (2026)
- Progress in Nuclear Energy (2026)
- Nuclear Technology (2026)
- IEEE Transactions on Systems, Man, and Cybernetics: Systems (2026)
- Scientific Reports (2026)

Conference Reviewer

- KNS Conference – Nuclear I&C, Human Factors, and Automated Remote Systems Division (2024 – 2026)
- KNS Conference – Nuclear Safety Division (2025)

Other Service

- 분과 자문위원, 가상원자로 개발 사업 (2025 – 2026)

TRAINING

1. APR1400 Simulator operating training by Korea Institute of Nuclear Safety (Daejeon, South Korea, Aug. 2022)
2. RELAP5, TRACE, RELAP/SCDAPSIM, MARS-KS training course (Barcelona, Spain, Jun. 2018)